Lab 2: Cache Emulator

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Implementation Details

For the Implementation of the Cache Emulator we choose to use C++. The cache.hpp file can be referenced to see the main features of the emulator. It contains a cache class that holds the main data structure that represents the cache. It is esentailly a 2-D array that can be thought of as a matrix of n-sets x m-blocks.

```
Block Datastructure

struct Block {
int tag;
bool valid;
chrono::time_point<chrono::steady_clock> timestamp;
int m;
}

Cache Datastructure

vector<vector<Block>> cache;
```

As shown above the we represented the blocks as a struct that contains the tag, valid bit, timestamp, and replacement value m. The timestamp is used for enforcing the least recently used (LRU) replacement policy. The value m is used as the hot cold value, this is used to enforced the RRIP replacement policy. The cache itself is represented as a vector of vectors. With this we can index the outer vector to access a particular set then index the inner vector to access a particular block.

Verification

In order to verify that our cache emulator was working correctly we created a short hand trace file, and computed the hits and misses manually.

Time	Address	Binary	Tag	Set	Set 0	Set 1	Eviction	Hit/Miss
T1	`0x10`	`0001 0 000`	1	0	[**1**,-]	[-,-]		Miss
T2	`0x20`	`0010 0 000`	2	0	[1,**2**]	[-,-]		Miss
T3	`0x10`	`0001 0 000`	1	0	[**1**,2]	[-,-]		**Hit**
T4	`0x30`	`0011 0 000`	3	0	[1,**3**]	[-,-]	Tag 2 (T2)	Miss
T5	`0x20`	`0010 0 000`	2	0	[**2**,3]	[-,-]	Tag 1 (T3)	Miss

T6	`0x30`	`0011 0 000`	3	0	[2,**3**]	[-,-]		**Hit**
T7	`0x48`	`0100 1 000`	4	1	[2,3]	[**4**,-]		Miss
T8	`0x30`	`0011 0 000`	3	0	[2,**3**]	[4,-]		**Hit**
T9	`0x18`	`0001 1 000`	1	1	[2,3]	[4,**1**]		Miss
T10	`0x18`	`0001 1 000`	1	1	[2,3]	[4,**1**]		**Hit**

As seen in the table the hand trace resulted in four hits and six misses with a total of ten memory accesses. The miss rate can be calculated by.

$$MissRate = \frac{\#Misses}{\#MemoryAccesses} = \frac{6}{10} = .60 = 60\% \text{ miss rate}$$

We can compare the results of our cache emulator to verify it is working properly. This can be done by running the following command.

```
.\cache_emulator --blocks 2 --sets 2  --size 8 --trace ./trace_files/verification.trace
```

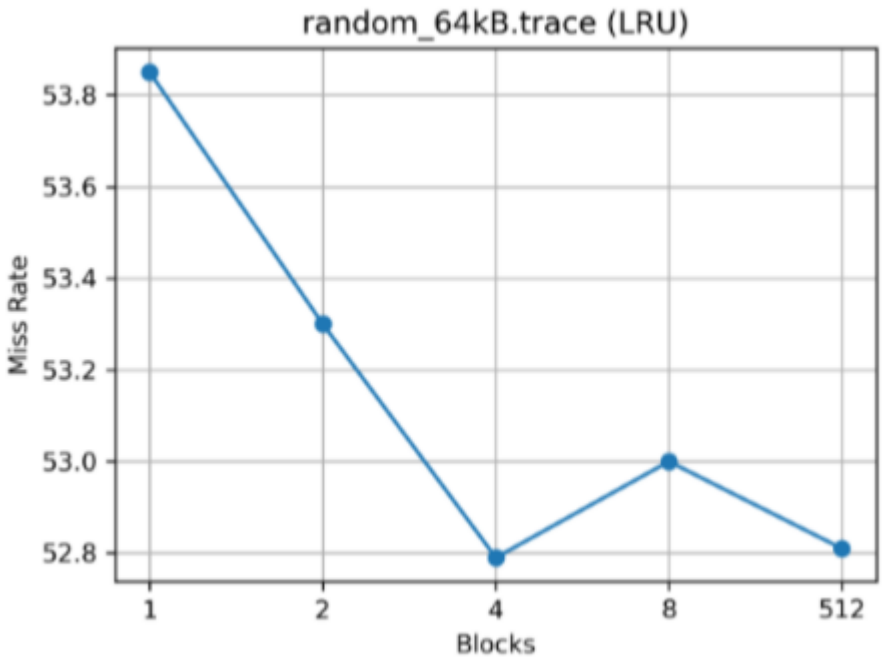
```
Cache-Emulator lab_report ● ? × ./cache_emulator --blocks 2 --sets 2  --size 8 --trace ./trace_files/verification.trace
num_sets: 2
num_blocks: 2
block_size: 8
filename: ./trace_files/verification.trace
RRIP: false

accesses: 10
num_hits: 4
num_misses: 6
Miss Rate: 60.00%
```

We can see with the output above we get the expected miss rate of 60%

Results

Experiment A: Associativity Sensitivity



Experiment B: The Comparison

Conclusion